The Risers: changes since image v3

PHAIL stage 1, board molmo.competition_v2. Work of 7 to 10 November 2026. Times are Central European Time.

Summary

- **Best board score: still image v3, 219/320 = 68.4%** (7 November). On 10 November we are 6th of 39 teams. 5th place has 225 (6 episodes more), 1st place has 238.
- Since v3 we made **5 more board submissions** (v4-push 217, v4 215, v3 again 218, v5 215, v5-touch 197), about **20 local test runs** and **1 cloud test run**, and we **trained our own action expert** for open and close tasks on a cloud H100.
- The gap to the top 5 teams is in **open tasks (6.8 episodes)** and **hard close tasks (4.8 episodes)**. On pick tasks we are level with them.
- **What worked on its target tasks:**
  – Close push rule: hard close tasks 4 → 9 of 15 on the board (image v4).
  – Camera switch when the left camera is blocked: +1 episode in the local test.
  – Touch grasp for open tasks: 4 → 6 of 30 in the local test, and 4 of 30 on the board (our best open result).
- **What did not work:**
  – Reword open tasks as "pick up the X handle": 0 of 30.
  – Our own open/close action expert alone: no gain, and slower on normal close tasks.
  – Regrasp rule: 4 of 30, no gain.
  – 10 flow steps for open and close tasks: 2 of 30.
  – MolmoAct2 for close tasks: does not fit the board GPU.
- **Why the total did not go up:**
  – One board run varies by about ±6 episodes, which is as large as the gains.
  – Images v5 and v5-touch move an action expert between the CPU and the GPU when the task kind changes. Each move takes about 290 ms. The board charges this time to the robot and runs about 4 episodes at once. So pick tasks fell: v5-touch solved 175 of 260 pick episodes, against 194 to 199 in the earlier runs.
- **Next:** image v6 = v5-touch with both action experts on the GPU. Test it with 4 episodes at a time before the submission. From 10 November there is 1 submission per day; stage 1 ends on 15 November.
- **Cloud cost since v3: $27.50** (demo generation $6.07, training $18.62, the 34-episode test $2.81).

Timeline

When	What we did	Where	Result
7 Nov 15:39	Submit v3 (3 flow steps, still-arm rule)	Board	219 (68.4%), rank 2 of 33
7 Nov evening	Camera check of the v3 replays	Analysis	Left camera blocked in about 7 episodes
7 Nov evening	Close push rule and camera switch: 15 hard close episodes	Local PC	Base 4, push 7, camera 6, image v4 7
7 Nov 22:37	Submit v4-push	Board	217 (67.8%), hard close 6
7 Nov 22:43	Submit v4 (push and camera switch)	Board	215 (67.2%), hard close 9
8 Nov 06:14	Submit v3 again	Board	218 (68.1%): the noise is about ±6
8 Nov	Study of open tasks (videos, joints, gripper data)	Analysis	The gripper grasps air; the model never saw open tasks
8 Nov	Open tests: base, reword, reword + pull, pull (30 open episodes)	Local PC	3, 0, 3, 4

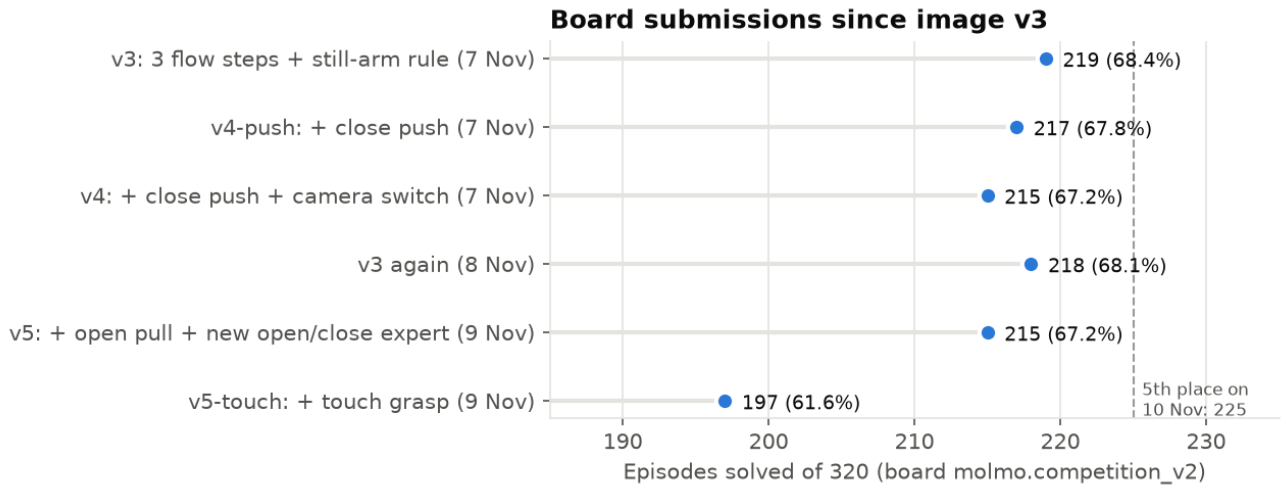
When	What we did	Where	Result
8 Nov	Study of 37 hard pick failures	Analysis	5 causes (table in the 8 November section)
8–9 Nov	Generate 1,955 open and close demos	Local PC and cloud L40S	$6.07 for the cloud part
9 Nov 11:02–13:27	Train our open/close action expert (2,500 steps)	Cloud H100	Loss 0.067 → 0.022, $18.62
9 Nov 13:38	Submit v5 (our expert, open pull)	Board	215 (67.2%), open 1
9 Nov	Saved experts, step 1,000 and step 2,400 (30 open episodes)	Local PC	2 and 4
9 Nov	Regrasp rule (30 open episodes)	Local PC	4
9 Nov	10 flow steps (30 open episodes)	Local PC	2
9 Nov	Touch grasp (30 open episodes)	Local PC	**6, the best**
9 Nov evening	Study of the top 5 replays; 34 "reliable" test episodes	Analysis	Gap: open 6.8, hard close 4.8
9 Nov 22:14–23:21	v3 and v5-touch on the 34 reliable episodes	Cloud L40S	15 and 18 of 34, $2.81
9 Nov 23:31	Submit v5-touch	Board	197 (61.6%): open 4, pick tasks fell
10 Nov	Find the cause of the pick fall	Analysis	Expert swap, 290 ms charged to the robot

Score progress (newest first)

Every board submission. "Change" compares the score with our best score before that submission.

Date (UTC)	Submission	Model and setting	Solved	Score	Change	Board rank
9 Nov	`fdf9b9845e06f7c`	v5-touch: v5 + touch grasp for open tasks	197	61.6%	−6.8	–
9 Nov	`3f52961f7f2dcab4`	v5: v4 + open pull + our open/close action expert	215	67.2%	−1.2	–
8 Nov	`2e6db64841627b1e`	v3 again (the same image)	218	68.1%	−0.3	–
7 Nov	`2e63188780383ff`	v4: v3 + close push + camera switch	215	67.2%	−1.2	–
7 Nov	`2518efd8af2927f6`	v4-push: v3 + close push	217	67.8%	−0.6	–
7 Nov	`2a74d6b52a67c4be`	**MolmoBot-Img, 3 flow steps, still-arm rule (image v3)**	**219**	**68.4%**	**+8.4**	**2 of 33**
6 Nov	`58f19b77fa0acea3`	MolmoBot-Img, 12 actions per call, 5 flow steps (image v2)	192	60.0%	+12.8	12 of 33
5 Nov	`30b01a00181a225`	MolmoBot-Img-DROID, half overlap, left camera (image v1)	151	47.2%	+6.6	13 of 33
5 Nov	`6571687c389055ca`	MolmoBot-DROID, half-overlap timing	130	40.6%	+4.0	14 of 33
5 Nov	`783e6095782a813c`	MolmoBot-DROID, normal timing	117	36.6%	+8.5	14 of 32
2 Nov	5 runs	MolmoAct2-DROID again, the same image 5 times	82–89	25.6–27.8%	−0.3 to −2.5	11 of 20
1 Nov	`28462ec4cc7fb2b5`	GR00T N1.7, our fine-tune (step 10000)	66	20.6%	−7.5	–
31 Oct	`2d30ed2cf955a93`	MolmoAct2-DROID with the GPU fix	90	28.1%	+8.7	10 of 15
31 Oct	`3208ed8ba648246a`	MolmoAct2-DROID, first image	–	error	–	–
31 Oct	`1dd6e85f68e0c3ab`	π0.5-DROID (baseline model)	62	19.4%	first	–

The board shows the best score of each team, so a worse submission does not lower it.



Successes by task type

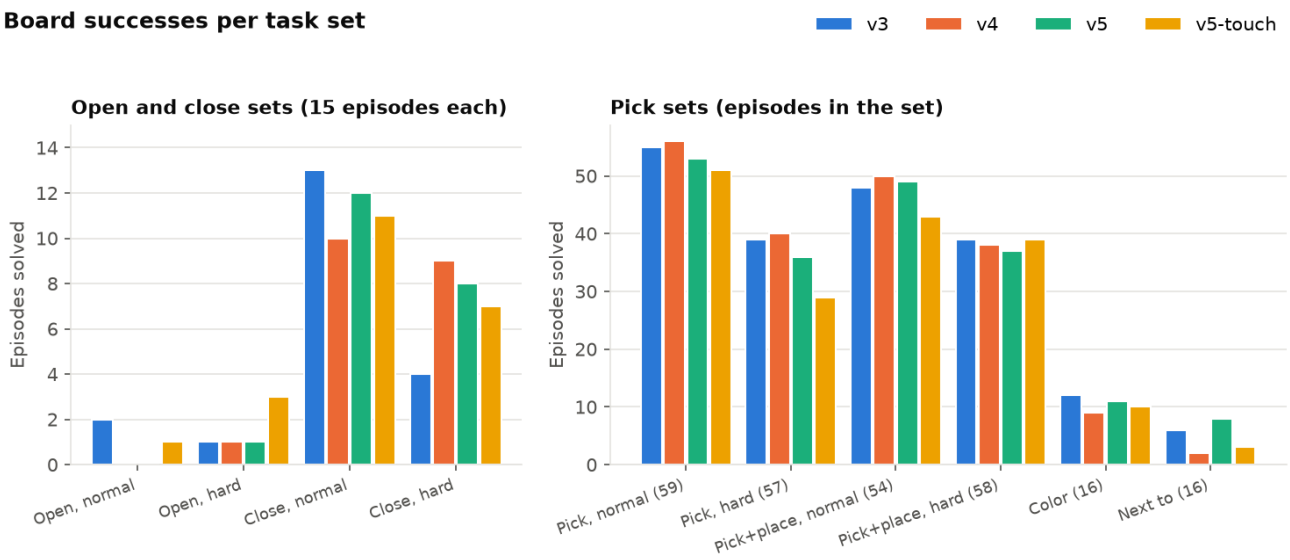
Task type	Episodes	v5-touch (9 Nov)	v5 (9 Nov)	v4 (7 Nov)	v4-push (7 Nov)	v3 (7 Nov)	v2 (6 Nov)	Img v1 (5 Nov)	MolmoBot (5 Nov)	MolmoAct2 (31 Oct)
Pick	116	80	89	**96**	91	94	85	70	55	28
Pick and place (with color)	128	92	97	97	98	**99**	84	65	48	32
Place next to	16	3	**8**	2	6	6	4	0	0	3

Task type	Episodes	v5-touch (9 Nov)	v5 (9 Nov)	v4 (7 Nov)	v4-push (7 Nov)	v3 (7 Nov)	v2 (6 Nov)	Img v1 (5 Nov)	MolmoBot (5 Nov)	MolmoAct2 (31 Oct)
Open	30	4	1	1	3	3	2	4	2	**6**
Close	30	18	20	19	19	17	17	12	12	**21**
Total	320	197	215	215	217	**219**	192	151	117	90

Successes by task set (MolmoBot submissions)

Set	Episodes	Time limit	v5-touch	v5	v4	v4-push	v3	v2	Img v1
Pick, normal rooms	59	31 s	51	53	**56**	54	55	**56**	**56**
Pick, hard	57	21 s	29	36	**40**	37	39	29	14
Pick and place, normal rooms	54	46 s	43	49	**50**	45	48	39	37
Pick and place, hard	58	41 s	39	37	38	**41**	39	34	19
Pick and place, color	16	41 s	10	11	9	**12**	**12**	11	9
Place next to	16	31 s	3	**8**	2	6	6	4	0
Open, data generation	15	31 s	1	0	0	2	2	1	**3**
Open, hard	15	31 s	**3**	1	1	1	1	1	1
Close, data generation	15	31 s	11	12	10	**13**	**13**	11	9
Close, hard	15	31 s	7	8	**9**	6	4	6	3
Total	320		197	215	215	217	**219**	192	151

`risers/competition_runs/board_by_set.py` makes these counts from the `result.json` file of each submission.



What helped and what did not

Changes that helped

Date	Change	Evidence	Board gain
31 Oct	MolmoAct2-DROID instead of π0.5	Board: 90 against 62	+8.7 points
5 Nov	MolmoBot-DROID instead of MolmoAct2	Local pick test 17/20 against 8/20	+8.5 points
5 Nov	Half-overlap timing (ask for the next actions while the robot moves)	Robot waits 69% instead of 73%	+4.0 points
5 Nov	MolmoBot-Img-DROID (1 frame, faster) and the left shoulder camera	Local pick test 18/20 against 16/20	+6.6 points

Date	Change	Evidence	Board gain
6 Nov	Play 12 of 16 actions per call, and 5 flow steps instead of 10	Local hard test 16/28 against 3/28	**+12.8 points**
7 Nov	Still-arm rule, and 3 flow steps instead of 5	Local frozen test 10/19 against 4/19	+8.4 points
7 Nov	Close push rule (close tasks)	Local hard close 7/15 against 4/15. Board hard close 6 and 9 against 4	Hard close +2 to +5. Total in the noise
7 Nov	Camera switch when the left camera is blocked	Local hard close 6/15 against 4/15 (#79 solved)	About +1
9 Nov	Touch grasp (open tasks)	Local open 6/30 against 4/30. Board open 4/30, our best	Open +1 to +3. Total fell (see 9 November)

Changes that did not help

Date	Change	Result
31 Oct	MolmoAct2 with a shorter action horizon (0.5 s)	Practice 5/20 against 8/20 with the default. Dropped
1 Nov	GR00T N1.7 fine-tuned on 1,577 of our own demos	20.6% on the board, below MolmoAct2 (28.1%) and below untrained GR00T (25.0%)
2 Nov	Submit the same MolmoAct2 image again	5 more runs: 25.6% to 27.8%. The best stayed 28.1%
5 Nov	Full-overlap timing (drop 8 of 16 actions)	Local 14/20 against 17/20 with normal timing
6 Nov	Gripper rule: a closed gripper opens only when the model value is below 40	Local 16/28, the same as without the rule
7 Nov	Give the model the wording of its training data	Frozen episodes 3/19 against 2/19: no effect
7 Nov	Ask the model again, use the other camera, or use 10 flow steps on a frozen arm	Probe: the arm moves at 0, 4 and 0 of 16 frozen moments
7 Nov	MolmoAct2 for close tasks beside MolmoBot	Not possible: 15 GB + 10.5 GB does not fit the 24 GB board GPU
8 Nov	Reword "open the X" to "pick up the X handle"	Local open 0/30 against 3/30. It removes all wins
8 Nov	Open pull (pull back when the gripper holds a handle)	Local open 4/30 against 3/30: +1, in the noise. The model almost never holds a handle
9 Nov	Our own open/close action expert (1,955 new demos, H100 training)	Local open 4/30 (= original expert). Board open 1/30, normal close slower
9 Nov	Regrasp rule (grasp again after a close on nothing)	Local open 4/30: no gain. 72 regrasps, 1 caught something
9 Nov	10 flow steps for open and close tasks	Local open 2/30
9 Nov	Two action experts that change places on the GPU	Each change costs about 290 ms of robot time on the board. Pick tasks fell in v5 and v5-touch

Run-to-run noise

The same behaviour gives different results in two board runs. v4-push and v3 act the same on 290 of the 320 episodes (all but the close tasks). On those episodes v4-push solved 27 that v3 did not and lost 31 that v3 solved. A second run of v3 gave 218 against 219. So one board run varies by about ±6 episodes (±2 points). A change must give more than that, or we cannot see it in one run.

9 November (evening): top-5 study, cloud test and image v5-touch (61.6%)

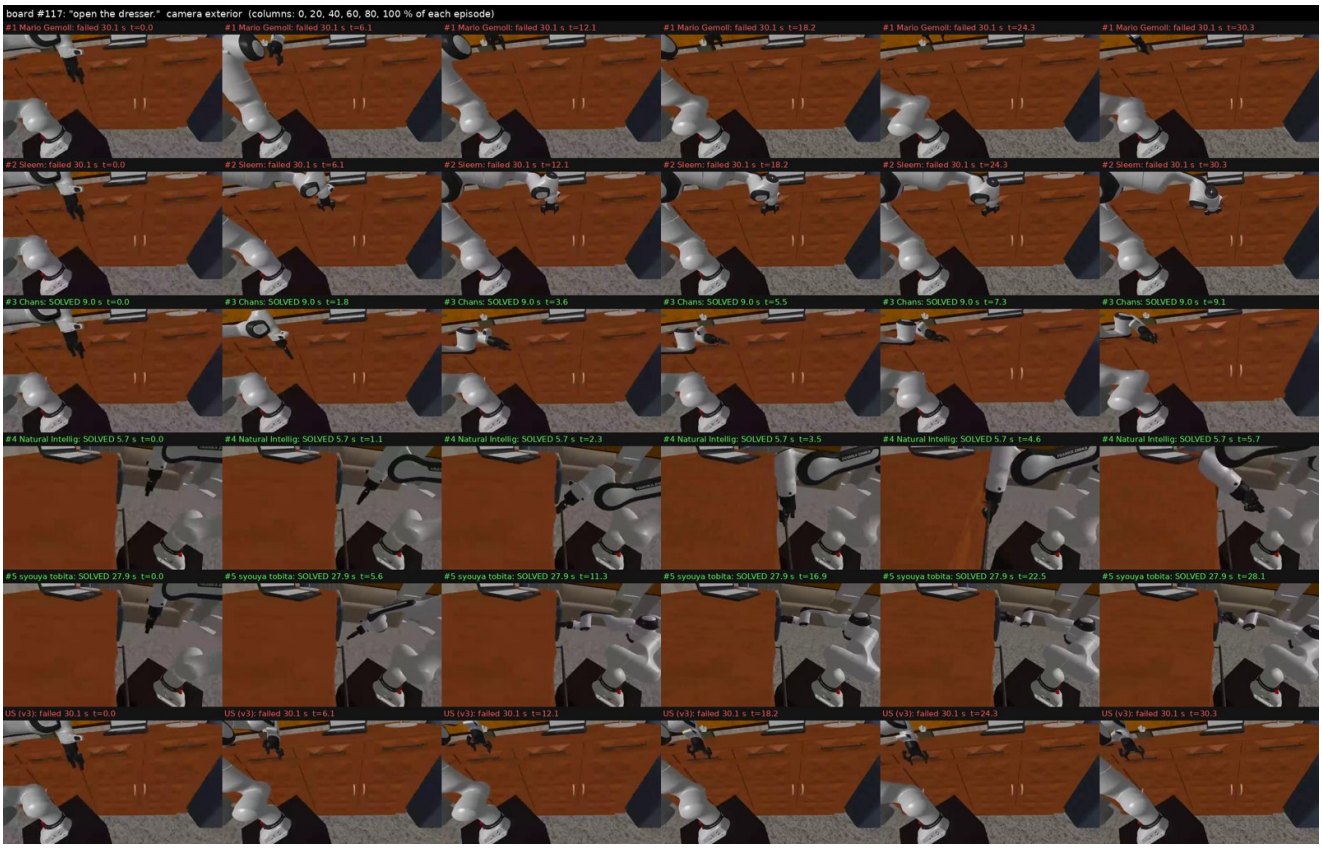
Study of the top 5 teams

The board shows a public replay (videos) of the best run of each team. The rules say: "Do not train on data from our evaluation runs, other than the results of your own team". We used the replays only to compare and to choose test episodes. We did not train on them.

`risers/competition_runs/board_videos/download_board.py` reads the board without a key and downloads the replays of the top 5 and our own. `compare_top.py` compares the successes per set:

Set	Episodes	#1	#2	#3	#4	#5	Our v3	Top-5 mean − v3
Pick, normal	59	56	57	55	56	55	55	+0.8
Pick and place, hard	58	41	42	35	40	39	39	+0.4
Pick, hard	57	44	39	37	33	41	39	−0.2
Pick and place, normal	54	49	45	44	47	43	48	−2.4
Pick and place, color	16	15	15	15	11	13	12	+1.8
Place next to	16	8	4	0	5	4	6	−1.8
Close, normal	15	13	13	15	14	11	13	+0.2
Open, normal	15	4	6	6	5	5	2	+3.2
Open, hard	15	4	5	6	4	4	1	+3.6
Close, hard	15	4	12	12	9	7	4	+4.8
All	320	238	238	225	224	222	219	**+10.4**

The gap to the top 5 is in the open tasks (+6.8 episodes) and the hard close tasks (+4.8). On pick tasks we are level with them. 8 of the 30 open episodes are solved by no top-5 team.



Episode #117 "open the dresser.", 6 frames of each team (exterior camera). Three of the top 5 open the drawer; #1, #2 and our v3 fail. The top teams face the drawer front early.

How their arms move. The replays have only videos, no gripper data. `grip_from_video.py` reads the gripper closure from the wrist video. It is a linear model of the finger area of the image, trained on our own local runs, where the true gripper signal is known. On local runs it did not see, it finds the first close within 1 s in 74% of the episodes. On the hard pick sets the wrist camera sits differently in each episode, and the reading is wrong in 29% to 50% of them. So we use only the 6 sets where it is right.

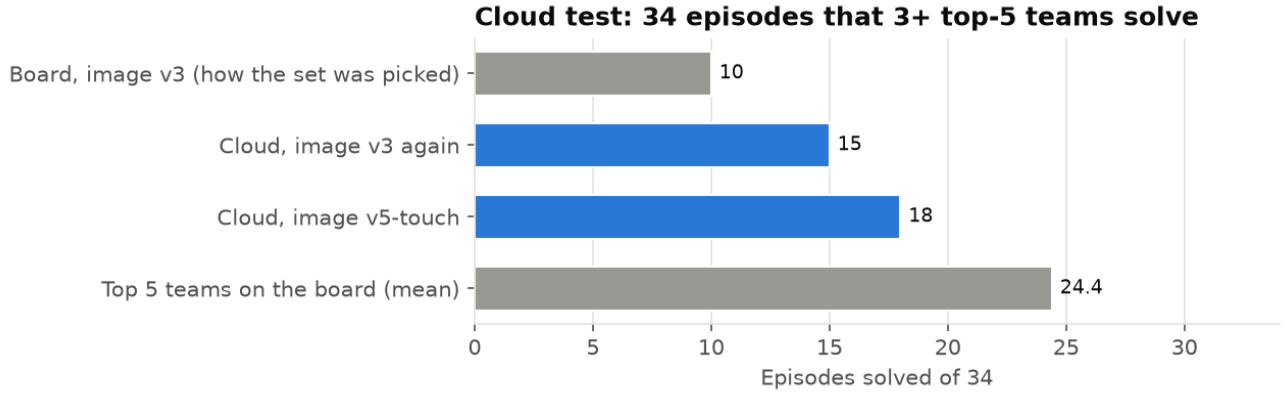
Set	Top-5 successes: first close, closes	Our v3 successes: first close, closes	Our v3 failures: first close, closes
Pick, normal	2.9 s, 1	3.4 s, 1	3.4 s, 10
Pick and place, normal	3.7 s, 1	3.8 s, 1	3.6 s, 11

Set	Top-5 successes: first close, closes	Our v3 successes: first close, closes	Our v3 failures: first close, closes
Open, normal	8.8 s, 1	11.7 s, 1	4.9 s, 9
Close, normal	3.0 s, 0 (push)	2.8 s, 0	5.3 s, 10

- When we succeed, we move like the top teams: one grasp at about the same time.
- When we fail, the gripper closes 9 to 11 times: the first grasp misses and every retry misses too.
- In open tasks the top teams close the gripper at about 8.8 s, after they reach the handle. Our failures close at about 4.9 s, before the handle. This supports the touch grasp (below).
- In close tasks the top teams push the door shut and do not close the gripper. v3 grasps again and again. The close push rule of v4 does the same as the top teams.

Test set of 34 "reliable" episodes, run in the cloud

`tests_1108/reliable34.json` holds the 34 board episodes that 3 or more of the top 5 solve and that 2 or more of our 3 board runs (v3, v4, v5) failed. We ran v3 and v5-touch on them on a Nebius cloud VM (L40S GPU, 32 CPUs). Three test streams ran side by side, each with its own policy server (a fourth server did not fit in the 48 GB GPU). Time 1.1 h, cost $2.53 (plus $0.28 for a first start that stopped at the setup).



Group	Episodes	Board v3	Cloud v3	Cloud v5-touch	Top-5 mean
Open and close	11	4	3	5	7.8
Pick and pick and place	23	6	12	13	16.6
All	34	10	15	**18**	24.4

v3 gets 15 in the second run because the set holds episodes that we failed before; the second v3 run agrees with the board run on only 21 of the 34 episodes. So the fair comparison is v5-touch against v3 in the same cloud run: 18 against 15. In open and close tasks v5-touch gains: hard close 0 → 2, normal open 2 → 3. It lost one normal close episode (#86), where v5 is slow.

Board result of v5-touch: 61.6%

We submitted v5-touch (v5 + touch grasp) on 9 November, while 5 submissions per day were still allowed.

	v3	v4	v5	v5-touch
Open (30)	3	1	1	**4**
Close (30)	17	19	**20**	18
Pick and pick and place (260)	**199**	195	194	175
Total	**219**	215	215	197

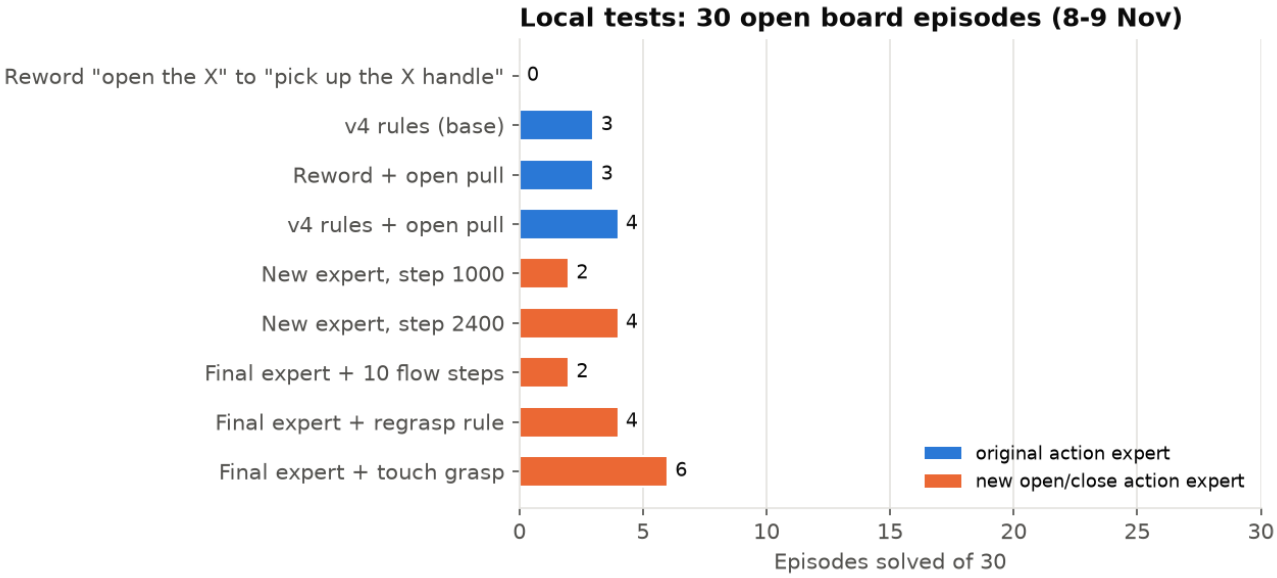
Open tasks gave our best board result (4/30). But pick tasks fell by about 20 episodes. In the 4 earlier runs with per-set data the pick tasks were always 194 to 199, so this is not noise. The pick code is the same in v5 and v5-touch (we compared the code inside the two images).

Cause (most likely): the action expert swap. Images v5 and v5-touch hold two action experts. Only the one in use stays on the GPU; the other waits on the CPU. The board runs about 4 episodes at a time on one policy server, so pick and open/close calls alternate. v5 changed the expert 1,814 times and v5-touch 1,509 times. A change moves 2 × 1.07 GB between the CPU and the GPU: **290 ms** on our PC (measured in the image), more than a model call (180 ms). The platform charges the robot the wall time of each model call (`eval.charge_inference_time = True`, also in the simulator), and the server does one call at a time. So each change also stops the other episodes. The hard pick set has the shortest time limit (21 s), and it lost the most: v3 39, v4 40, v5 36, v5-touch 29. Our local and cloud tests run one episode per server, so they did not show this.

Fix: keep both action experts on the GPU (1.07 GB more GPU memory). The next image will have this fix.

9 November: grasp rules for open tasks (local tests)

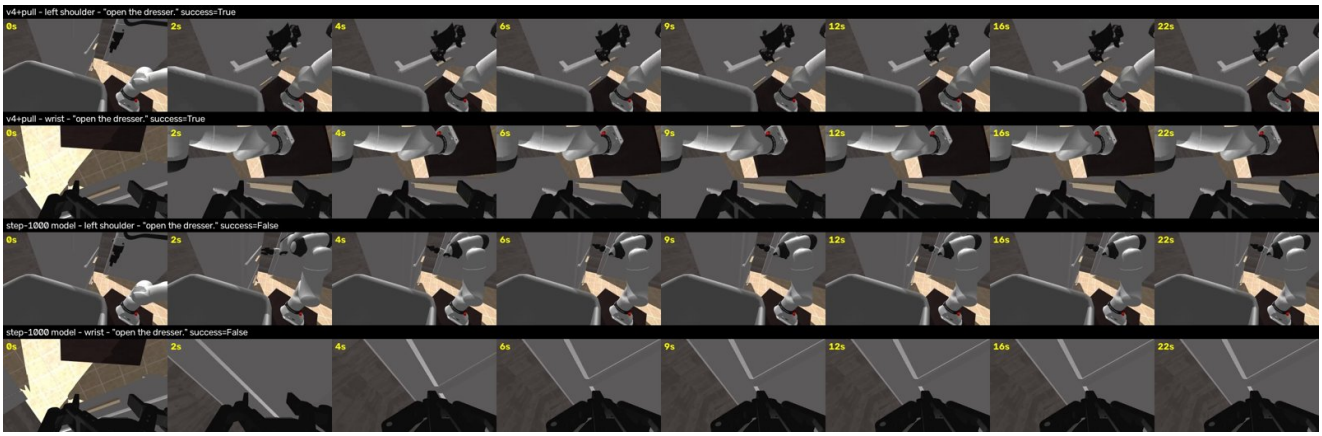
All tests on the 30 open episodes of the board (`tests_1108/open30.json`), on the local PC, one episode at a time.



Run	Action expert	Rules	Open, normal	Open, hard	Solved
op_base	original	v4 rules	2/15	1/15	3
op_word	original	v4 + reword to "pick up the X handle"	0/15	0/15	0
op_both2	original	v4 + reword + open pull	3/15	0/15	3
op_pull	original	v4 + open pull	3/15	1/15	4
op_step1000	ours, step 1000	v4 + open pull	1/15	1/15	2
op_step2400	ours, step 2400	v4 + open pull	3/15	1/15	4
op_regrasp	ours, final	v4 + open pull + regrasp	3/15	1/15	4
op_flow10	ours, final	v4 + open pull + 10 flow steps	1/15	1/15	2
op_touch	ours, final	v4 + open pull + touch grasp	**5/15**	1/15	**6**

Why the model does not open: it does not hold the handle

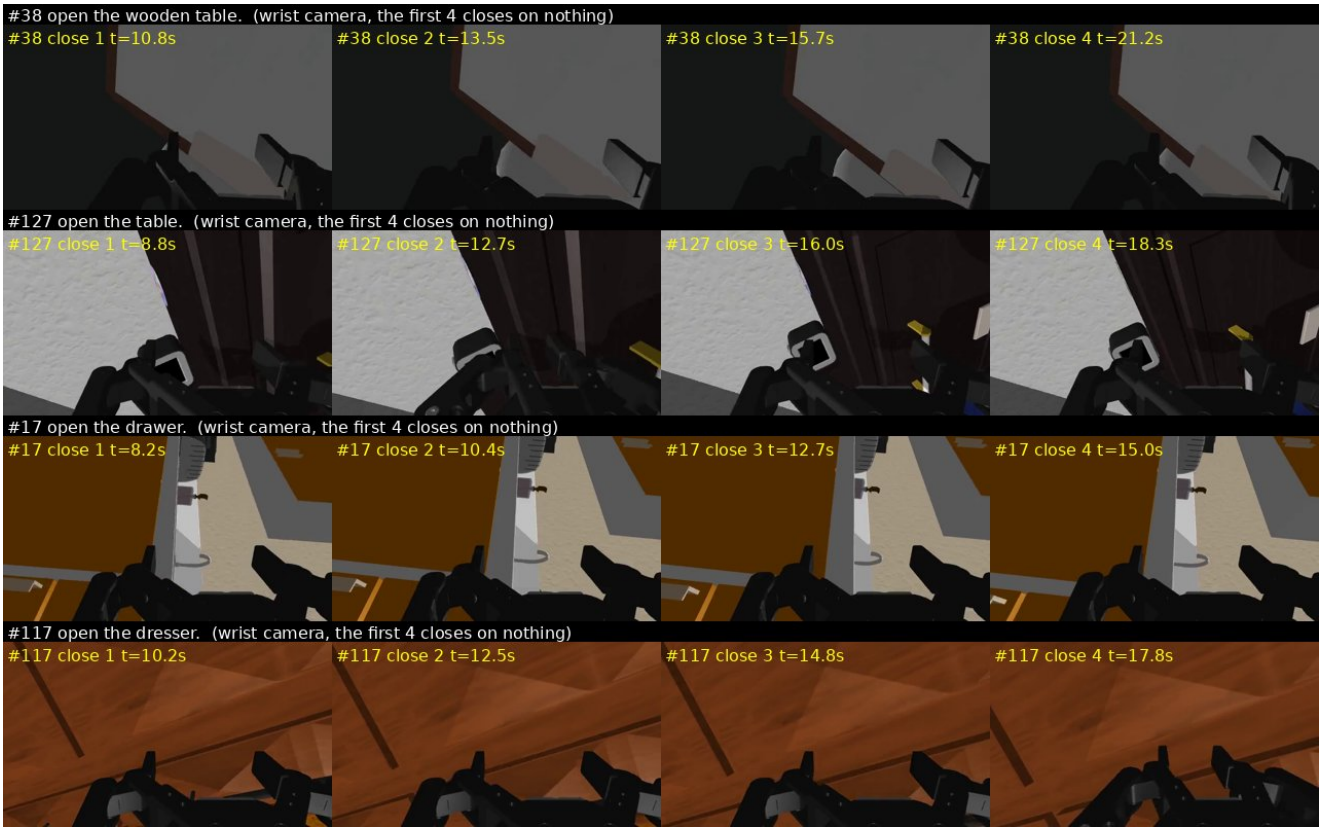
The gripper data of the local runs shows that neither expert holds a handle. The longest hold in an episode is 0.5 to 0.7 s (median), and the hand moves about 1 cm while closed. Our demos close once at about 5 s, hold about 5 s and pull 16 to 21 cm. Most open wins of both experts came from a bump of the arm, not from a grasp and a pull.



Episode #107 "open the dresser.": the original expert with the open pull (top two rows) opens the drawer. The step-1000 expert (bottom two rows) moves to the side of the drawer and stays there.

Regrasp rule: no gain

Option `--model.open_regrasp`: when a level gripper closes on nothing, it backs off 3 cm, moves 6 cm in with an offset of 0, +3 or −3 cm along the finger axis, and closes again. Result 4/30, the same as without it. The rule fired 72 times in 20 episodes, and only 1 try closed on something. Each try costs about 2.3 s of the 31 s. The wrist pictures show why: the handle is 5 to 15 cm away or not in view. The misses are aim errors, not small offsets.



Wrist camera at the first 4 closes of 4 failed open episodes: each close is on nothing.

10 flow steps for open and close tasks: worse

Option `--model.open_close_flow_steps=10` (the model default; we use 3). Result 2/30.

Touch grasp: the best open result (6/30)

Option `--model.open_touch`. When the model closes a level, open gripper in an open task, a script takes over:

1. On a drawer-like task (drawer, dresser, desk, table, nightstand, chest), turn the fingers so they close vertically (a drawer handle is mostly a horizontal bar).
2. Move the open gripper in along its axis, 3 cm per call, until a call moves it less than 1 cm (it touches something) or after 3 calls.
3. Close. When two readings in a row show the fingers stopped on something, pull (the open pull, chained).

At most 2 touch grasps per episode. Result: 6/30 (normal 5/15, hard 1/15). 56 touch grasps, and 13 pulls in 12 episodes; the earlier runs had 0 to 8 pulls.

9 November: open/close data, cloud training and image v5 (67.2%)

Why train: the model never saw open or close tasks

The MolmoBot training data has 8 task kinds on 2 robots. For the Franka arm there are only pick, pick and place, color and next-to tasks. Open and door data exists only for the RB-Y1 robot. So the DROID model never saw open or close tasks.

Data generation

`risers/datagen/risers_datagen.py` adds MolmoSpaces data configs that make open and close demos with the scripted planner of MolmoSpaces:

- `RisersOpenMore`, `RisersCloseMore`: normal rooms (iTHOR train rooms; the board uses the validation rooms).
- `RisersOpenHard*`, `RisersCloseHard*` (iTHOR, ProcTHOR, Holodeck): copies of the board's hard benchmark settings (start pose noise 0.26 rad, height ±0.25 m, base turn ±0.52 rad) with the board's hard cameras.

The local PC made the normal demos (3 workers). A Nebius cloud VM (L40S, 16 workers) made the hard demos in 2.7 hours ($6.07).



Left: board episodes. Right: our generated demos of the same task kinds.

Demos	Normal rooms	Hard settings	All
Open	668	574	1,242
Close	229	484	713
All	897	1,058	**1,955**

Each demo is used twice in training, once with the left and once with the right shoulder camera (3,910 trajectories).

Training on the H100

Model	MolmoBot-Img-DROID. Only the action expert trains (536 M parameters); the rest is frozen
Steps	2,500 steps × 128 trajectories, about 3.5 s per step (2 h 25 min)
Machine	Nebius H100 80 GB VM, 4.1 h with setup, $18.62
Saves	Every 200 steps, copied to the local disk (the first run stopped at step 1,160 to add the saves; the restart gave the same loss at every step)
Loss	Action flow loss 0.067 (step 20) → 0.022 (step 2,500)



Image v5: two action experts

New server options `--model.open_close_expert=<file>` and `--model.open_close_expert_tasks=both|open|close`. Tasks that start with "open the" or "close the" use our action expert; all other tasks use the original one. Image v5 = v4 flags + `--model.open_pull=True` + our expert (patch `0010`).

Board result (submission `3f52961f7f2dcab4`): **215/320 = 67.2%**. Open 1/30, hard close 8/15 (v3: 4), normal close 12/15 with a median time of 7.8 s (v3: 1.9 s). Our expert levels the gripper only after a median of 6.4 to 7.0 s; the demos do it in 2.8 s.

Local tests of the saved experts

Step 1,000: 2/30 open. Step 2,400: 4/30, the same as the original expert with the open pull. We stopped the tests of steps 2,000, 1,400 and 800 because the later steps gave no gain.

8 November: open tasks, the open pull and v3 again (68.1%)

v3 again: 218

The same v3 image gave 218 (first run 219). This measures the noise (see "Run-to-run noise" above).

Why open tasks fail

- All 30 open episodes start closed. Success: the joint opens to 15% of its range or more. 47 of the 60 open and close objects slide (drawers), 13 turn on a hinge (doors).
- In failed open episodes of v4 the gripper travels a median of 2.0 m in 30 s, but never more than 39 cm from its start. It wanders back and forth. The fingers are closed 35% to 40% of the time, and 90% of those closed readings are fully shut: the gripper grasps air.
- Gap to the 1st team: all in the normal sets (open normal 1.3 against 5, close normal 12 against 15). On the hard sets our mean is equal to theirs.

Open matrix

Two new options (patch `0010`): `--model.open_as_pick` (the open task becomes "pick up the X handle.") and `--model.open_pull` (when the gripper holds something, pull it straight out 6 mm per action for 2 calls, at most 3 times). Results: see the open test table above (op_base 3, op_word 0, op_both2 3, op_pull 4). Reword hurts; the pull gives +1.

Hard pick failures (v4 board, 37 failures)

Cause	Episodes
The object slipped after the grasp	12
Never near the target object	7
Held another object	7
Closed at the target, but every try missed	8
Object too large (toilet, pan, pot)	3

Of the 115 hard pick episodes, 59 were solved in all 3 runs (v3, v4-push, v4), 37 in some and 19 in none.

7 November (evening): close push and camera switch, images v4-push and v4 (67.8%, 67.2%)

Blocked left camera

`competition_runs/video_tools/camera_check.py` measures the detail (mean grey gradient) of each shoulder camera. In about 7 board episodes of v3 the left camera sees only furniture or a wall. Almost all are open or close tasks.



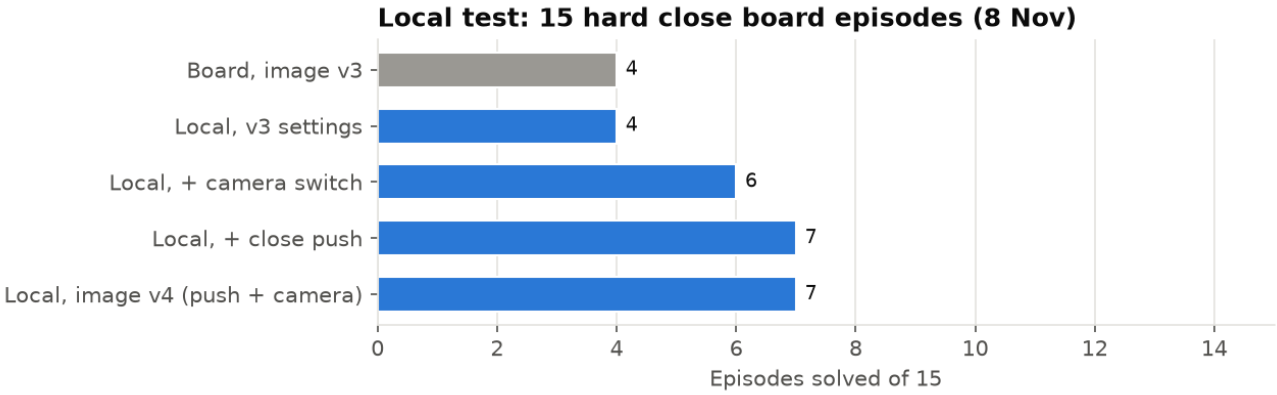
Left image of each pair: the left shoulder camera (blocked). Right image: the other shoulder camera.

Option `--model.camera_fallback=True`: when the detail of the left camera is below 0.6 and the other camera has 1.5 times more, the model uses the other camera.

Close push

The v3 videos show that in close tasks the robot reaches the door and then waits near it, or grasps it again and again. Option `--model.close_push_after=6` (patches `0007`, `0008`, with a numpy forward kinematics of the FR3 in `arm.py`): after 6 model calls, when the gripper stays within 8 cm for 4 readings, a script pushes. A gripper that points down drops 6 or 10 cm and then pushes 3 × 12.8 cm away from the robot base. A level gripper pushes along its axis. At most 3 pushes, with 4 model calls between them.

Local tests: 15 hard close episodes



Run	Hard close	Notes
v3 settings	4/15	The same 4 as the board
+ close push	7/15	Push solved #109 and #129 (and #149); faster #19 (27 → 11 s) and #49 (14.5 → 8.4 s). No losses
+ camera switch	6/15	Switch solved #79
image v4 (both)	7/15	The wins of both, no losses

Board results

Submission	Close, hard	Total
v3	4	219
v4-push (`2518efd8af2927f6`)	6	217 (67.8%)
v4: push + camera (`2e63188780383ff`)	9	215 (67.2%)

The fix works on the board: hard close went from 4 to 9. But the other 290 episodes vary by about ±6 per run, so the total did not go up.

We also checked MolmoAct2 for close tasks (it solved 8 or 9 hard close episodes on the board). MolmoAct2 (15 GB) and MolmoBot (10.5 GB) do not fit the 24 GB board GPU together, so we dropped it.

Cloud work and costs

All cloud work ran on Nebius virtual machines (VMs). Each VM was deleted at the end of its job.

Date	VM	GPU	Job	Time	Cost
9 Nov 07:11–09:50	risers-datagen	L40S 48 GB, 32 CPUs	Hard open and close demos (16 workers)	2.66 h	$6.07

Date	VM	GPU	Job	Time	Cost
9 Nov 09:29–13:38	risers-train	H100 80 GB, 16 CPUs	Train the action expert (a first run, then the final run with saves every 200 steps)	4.14 h	$18.62
9 Nov 22:05–22:12	risers-eval (try 1)	L40S 48 GB	Stopped at the setup (the VM's Python had no C headers); deleted at once	0.12 h	$0.28
9 Nov 22:14–23:21	risers-eval (try 2)	L40S 48 GB	v3 and v5-touch on the 34 reliable episodes, 3 streams side by side	1.11 h	$2.53
All				**8.0 h**	**$27.50**

Notes on the cloud test of 9 November:

- One policy server needs 10.9 GB of GPU memory. Three servers and their simulators fit in the 48 GB of the L40S. A fourth server stopped with "CUDA out of memory", so its 17 episodes ran in the free places of the other streams.
- One episode took 3 to 4.5 minutes per stream, about 1.5 times faster than the local PC. The local PC has only 15 GB of memory, and the policy server uses 7.8 GB of it, so the PC moves memory to the disk during the tests.
- Each stream had its own policy server. So the test did not show the expert swap cost of the board (about 4 episodes on one server).

Next steps

1. **Image v6:** v5-touch with both action experts on the GPU (no swap). The GPU has room: one server uses 10.9 GB of the 24 GB.
2. **Test as the board runs:** about 4 episodes at a time on one policy server. Compare v3 and v6 on the same episodes.
3. **Close tasks:** our action expert is slow on normal close tasks. Option: use it only for open tasks, and use the original expert with the close push for close tasks (v4 had 9 of 15 hard close).
4. **One submission per day** until 15 November. Submit only a change that is clearly better than the noise of about ±6 episodes.

Where the files are

What	Where
This report and its pictures	`reports_1110/` (project folder); on GitHub in `reports/`
Board results of each submission	`competition_runs/molmobot_img_v3`, `_v4`, `_v4_push`, `_v5_openclose`, `_v5_touch`
Local test tables	`tests_1108/results.md` (close), `results_open.md`, `results_checkpoints.md`, `results_reliable.md`, `results_reliable_touch.md`
Local test episodes	`~/evals/t8_*` on the PC; copies on the SSD in `local_tests_with_rules/` and `local_tests_reliable34/`
Trained action experts (every 200 steps and the final one)	SSD: `molmobot_openclose_2026-11-09/experts/` and `action_expert_open_close_final.pt`
Training logs and settings	SSD: `molmobot_openclose_2026-11-09/train.log`, `train_config.yaml`
Generated demos	SSD: `datagen/output/`
Test images (Docker)	Docker Hub `denistonantony/molmobot-img-droid` (tags v3, v4, v4-push, v5-openclose, v5-touch); the other test images on the SSD in `docker_images/`
Top-5 board replays	SSD: `board_replays/` (used only to compare, not to train)
Code changes	Positronic commits `7d0cf9cc`, `ce78d3e5`, `b48d55b2`, `794fd5e6` (local); patches `0007`–`0010` on GitHub